

Laixi Shi

3400 N. Charles St., Baltimore, Maryland 21218

✉ Email: laixis@jhu.edu

🌐 Website: <https://laixishi.github.io/>

Research Interests

Human-centered decision making, especially robust and data-efficient reinforcement learning ranging from foundation to applications, situated at the intersection of data science, optimization, and statistics.

Academic Appointments

2025.7 – Pre **Johns Hopkins University, MD, USA,**

Assistant Professor,

Department of Electrical and Computer Engineering & Data Science and AI Institute.

2023.8 – 2025.7 **California Institute of Technology, CA, USA,**

Postdoctoral Fellow, Computing + Mathematical Sciences,

Hosts: Adam Wierman and Eric Mazumdar.

Education

2018.8 – 2023.8 **Carnegie Mellon University, PA, USA,**

Ph.D. in Electrical and Computer Engineering,

Advisor: Yuejie Chi.

2014.9 – 2018.7 **Tsinghua University, Beijing, China,**

B.Eng., Electronic Engineering.

Work and Academia Experience

2022.9-12 **Google Research, Brain Team, Mountain View, United States,**

Student Researcher, Mentor: Pablo Samuel Castro, Matthieu Geist, Robert Dadashi.

2022.6-9 **Google Research, Brain Team, Paris, France,**

Research Intern, Mentor: Matthieu Geist, Robert Dadashi.

2020.5-8 **Mitsubishi Electric Research Laboratories (MERL), Boston, United States,**

Research Intern, Mentor: Dehong Liu.

2017.6-9 **Columbia University, New York, USA,**

Undergraduate Intern, Mentor: Xiaofan (Fred), Jiang.

2017.3-6 **Momenta, Beijing, China,**

Software Engineering Intern, Mentor: Gang Sun, Le Shan.

Selected Honors & Awards

2024 CMU ECE A.G. Milnes Award — the PhD thesis work judged to be of the highest quality and which has had or is likely to have significant impact in his or her field

2024 The Junior Researcher Workshop Invitation by Cornell

2024 Rising Stars In Electrical Engineering and Computer Science (EECS)

2024 The ISyE Junior Researcher Workshop Invitation by Georgia Tech

2023 Rising Stars In Machine Learning by University of Maryland

2023 Computing, Data, and Society Postdoctoral Fellow by Caltech

2023 ICASSP Rising Stars in Signal Processing
 2023 UT Austin Rising Stars in Computational and Data Sciences
 2022 UChicago Rising Stars in Data Science
 2022 NeurIPS 2022 Top Reviewer
 2022 Wei Shen and Xuehong Zhang Presidential Fellowship
 2019 Women in Machine Learning Scholarship
 2018 Presidential Fellowship granted by Carnegie Mellon University
 2018 Excellent Honors Graduate granted by Tsinghua University
 2017 The First Prize in 35th Tsinghua University Academic Challenge Cup
 2016 National Scholarship granted by the government of China
 2014 The Silver Medal of Chinese Physics Olympiad

Journals

- [1] [Laixi Shi](#), Gen Li, Yuting Wei, Yuxin Chen, Matthieu Geist, Yuejie Chi. "The Curious Price of Distributional Robustness in Reinforcement Learning with a Generative Model" *accepted by Operations Research*, 2026.
- [2] [Laixi Shi](#), Yuejie Chi. "Distributionally Robust Model-Based Offline Reinforcement Learning with Near-Optimal Sample Complexity" *Journal of Machine Learning Research (JMLR)*, 2024.
- [3] Gen Li, [Laixi Shi](#), Yuxin Chen, Yuejie Chi, Yuting Wei. "Settling the Sample Complexity of Model-Based Offline Reinforcement Learning" *The Annals of Statistics*, 2024.
- [4] Gen Li, [Laixi Shi](#), Yuxin Chen, Yuejie Chi. "Breaking the Sample Complexity Barrier to Regret-Optimal Model-Free Reinforcement Learning" *Information and Inference: A Journal of the IMA*, 2023.
- [5] [Laixi Shi](#) and Yuejie Chi. "Manifold Gradient Descent Solves Multi-channel Sparse Blind Deconvolution Provably and Efficiently." *IEEE Transactions on Information Theory*, 2021

Full Publications (Conferences and Workshops)

* represents equal contribution or alphabetical order.

Reinforcement Learning Theory: Reconcile Robustness with Data Efficiency

- [1] As, Yarden, Chengrui Qu, Benjamin Unger, Dongho Kang, Max van der Hart, [Laixi Shi](#), Stelian Coros, Adam Wierman, and Andreas Krause. "SPiDR: A Simple Approach for Zero-Shot Safety in Sim-to-Real Transfer." *Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
- [2] [Laixi Shi](#)*, Jingchu Gai*, Eric Mazumdar, Yuejie Chi, Adam Wierman. "Breaking the Curse of Multiagency in Robust Multi-Agent Reinforcement Learning" *International Conference on Machine Learning (ICML)*, 2025.
- [3] Chenbei Lu, [Laixi Shi](#), Zaiwei Chen, Chenye Wu, Adam Wierman. "Overcoming the Curse of Dimensionality in Reinforcement Learning Through Approximate Factorization" *International Conference on Machine Learning (ICML)*, 2025.
- [4] Chengrui Qu, [Laixi Shi](#), Kishan Panaganti, Pengcheng You, and Adam Wierman. "Hybrid Transfer Reinforcement Learning: Provable Sample Efficiency From Shifted-dynamics Data" *International Conference on Artificial Intelligence and Statistics*, 2025 (**AISTATS Oral**).
- [5] Eric Mazumdar*, Kishan Panaganti*, [Laixi Shi](#)*. "Behavioral Economics-Inspired Approach to Tractable Multi-Agent Reinforcement Learning" *International Conference on Learning Representations (ICLR Oral)*, 2025.

- [6] [Laixi Shi](#), Eric Mazumdar, Yuejie Chi, Adam Wierman. "Sample-Efficient Robust Multi-Agent Reinforcement Learning in the Face of Environmental Uncertainty" *International Conference on Machine Learning (ICML)*, 2024.
- [7] Pierre Clavier, [Laixi Shi](#), Erwan Le Pennec, Eric Mazumdar, Adam Wierman, Matthieu Geist. "Near-Optimal Distributionally Robust RL with General L_p Norms" *Conference on Neural Information Processing Systems (NeurIPS)*, 2024.
- [8] Jiin Woo, [Laixi Shi](#), Gauri Joshi, Yuejie Chi. "Federated Offline Reinforcement Learning: Collaborative Single-Policy Coverage Suffices" *International Conference on Machine Learning (ICML)*, 2024.
- [9] He Wang, [Laixi Shi](#), Yuejie Chi. "Sample Complexity of Offline Distributionally Robust Linear Markov Decision Processes" *Reinforcement Learning Conference (RLC)*, 2024.
- [10] [Laixi Shi](#), Gen Li, Yuting Wei, Yuxin Chen, Matthieu Geist, Yuejie Chi. "The Curious Price of Distributional Robustness in Reinforcement Learning with a Generative Model" *Conference on Neural Information Processing Systems (NeurIPS)*, 2023.
- [11] [Laixi Shi](#), Gen Li, Yuxin Chen, Yuting Wei, Yuejie Chi. "Pessimistic Q-Learning for Offline Reinforcement Learning: Towards Optimal Sample Complexity" *International Conference on Machine Learning (ICML)*, 2022.
- [12] Gen Li, [Laixi Shi](#), Yuxin Chen, Yuejie Chi. "Breaking the Sample Complexity Barrier to Regret-Optimal Model-Free Reinforcement Learning" *A short version at Conference on Neural Information Processing Systems (NeurIPS)*, 2021 (**NeurIPS Spotlight**).

Reinforcement Learning Practice: Control, Robotics, Resource Allocation

- [13] Shangding Gu*, [Laixi Shi*](#), Muning Wen, Ming Jin, Eric Mazumdar, Yuejie Chi, Adam Wierman, Costas Spanos. "Robust Gymnasium: A Unified Modular Benchmark for Robust Reinforcement Learning" *International Conference on Learning Representations (ICLR)*, 2025.
- [14] Yarden As, Chengrui Qu, Benjamin Unger, Dongho Kang, Max van der Hart, [Laixi Shi](#), Stelian Coros, Adam Wierman, Andreas Krause. "SPiDR: A Simple Approach for Zero-Shot Safety in Sim-to-Real Transfer" *Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
- [15] Shangding Gu*, [Laixi Shi*](#), Yuhao Ding, Alois Knoll, Costas Spanos, Adam Wierman, Ming Jin. "Enhancing Efficiency of Safe Reinforcement Learning via Sample Manipulation" *Conference on Neural Information Processing Systems (NeurIPS)*, 2024.
- [16] Haohong Lin, Wenhao Ding, Jian Chen, [Laixi Shi](#), Jiacheng Zhu, Bo Li, Ding Zhao. "BE-CAUSE: Bilinear Causal Representation for Generalizable Offline Model-based Reinforcement Learning" *Conference on Neural Information Processing Systems (NeurIPS)*, 2024.
- [17] [Laixi Shi](#), Robert Dadashi, Yuejie Chi, Pablo Samuel Castro, Matthieu Geist. "Offline Reinforcement Learning with On-Policy Q-Function Regularization" *European Conference on Machine Learning (ECML)*, 2023.
- [18] Wenhao Ding*, [Laixi Shi*](#), Yuejie Chi, Ding Zhao. "Seeing is not Believing: Robust Reinforcement Learning against Spurious Correlation" *Conference on Neural Information Processing Systems (NeurIPS)*, 2023.
- [19] Yiqi Wang, Mengdi Xu, [Laixi Shi](#), Yuejie Chi. "A Trajectory is Worth Three Sentences: Multimodal Transformer for Offline Reinforcement Learning" *The Conference on Uncertainty in Artificial Intelligence (UAI)*, 2023.
- [20] Peide Huang, Mengdi Xu, Jiacheng Zhu, [Laixi Shi](#), Fei Fang, Ding Zhao. "Curriculum Reinforcement Learning using Optimal Transport via Gradual Domain Adaptation" *Conference on Neural Information Processing Systems (NeurIPS)*, 2022.

Data-Driven Optimization for Inverse Problems: Sensing and Processing

- [21] [Laixi Shi](#), Dehong Liu, Jay Thornton. "Robust Camera Pose Estimation For Image Stitching" *IEEE International Conference on Image Processing (ICIP)*, 2021.
- [22] [Laixi Shi](#) and Yuejie Chi. "Manifold Gradient Descent Solves Multi-channel Sparse Blind Deconvolution Provably and Efficiently." *IEEE Transactions on Information Theory*, vol. 67, no. 7, pp. 4784-4811, 2021. Short version at *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2020.
- [23] T. Low, Y. Chi, J. Hoe, S. Kumar, A. Prabhakara, [L. Shi](#), U. Sridhar, N. Tukanov, C. Wang, and Y. Wu. "Zoom Out: Abstractions for Efficient Radar Algorithms on COTS Architectures." *IEEE International Symposium on Phased Array Systems and Technology (PAST)*, 2022.
- [24] [Laixi Shi](#), Dehong Liu, Masaki Umeda, and Norihiko Hana. "Fusion-Based Digital Image Correlation Framework for Strain Measurement." *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2021. Another version at *U.S. Patent Application 17/148,609*, 2022.
- [25] [Laixi Shi](#), Mostafa Mirshekari, Jonathon Fagert, Yuejie Chi, Hae Young Noh, Pei Zhang, and Shijia Pan. "Device-free Multiple People Localization through Floor Vibration." *First ACM Workshop on Device-Free Human Sensing*, 2019.
- [26] Sang Yu, [Laixi Shi](#), and Yimin Liu. "Micro hand gesture recognition system using ultrasonic active sensing." *IEEE Access*, vol. 6, pp. 49339-49347, 2018.

Patents

- [1] Dehong Liu, [Laixi Shi](#). "System and Method of Image Stitching using Robust Camera Pose Estimation." *US Patent App. 17/214,813*, 2022.
- [2] Dehong Liu, [Laixi Shi](#), Masaki Umeda, and Norihiko Hana. "Fusion-Based Digital Image Correlation Framework for Strain Measurement" *US Patent App. 17/148,609*, 2022.

Teaching Experience

- 2020 Spring 18202 Mathematical Foundations of Electrical Engineering, Grad TA, CMU ECE Department.
- 2021 Spring 18202 Mathematical Foundations of Electrical Engineering, Grad TA, CMU ECE Department.
- 2024 Summer Reinforcement Learning Bootcamp, Instructor, Caltech.

Current Students/Collaborators

- JHU: Juyang Bai (JHU ECE Ph.D. student), 2026-Present
- JHU & Caltech: Yashaswini Murthy, Postdoc co-advised with Prof. Adam Wierman, 2025-2026

Previous Mentorship

- Jingchu Gai (undergraduate student → CMU Ph.D. student), 2024-Present
- Yuchen Wu (master student → Ph.D. student at University of Washington), 2021-2022
- Gore Kao (master student → Tiktok), 2022-2023
- Yiqi Wang (master student → master in Robotics Institute at CMU), 2022-2024
- Chenbei Lu (visiting Ph.D. student → Postdoc at Cornell University), 2023-Present
- Zhengfei Zhang (visiting undergraduate student → Startup founder), 2023-2024

Selected Services and Outreach

- Area Chair: Conference on Parsimony and Learning (CPAL), 2024-2026.
8th Annual Learning for Dynamics & Control Conference (L4DC), 2026.
- Organization: • Breakout session leader at *3rd Women in Machine Learning Un-Workshop, ICML 2022*.
• Session Chair for the strategic and distributionally robust sequential decision making session at the 2024 INFORMS annual meeting, APS cluster.
- Journal Review: The Journal of Machine Learning Research, the IEEE Transactions on Information Theory, the IEEE Transactions on Signal Processing, Foundations of Computational Mathematics, Operations Research, Mathematics of Operations Research, the IEEE Transactions on Pattern Analysis and Machine Intelligence, IEEE Transactions on Control of Network Systems.
- Conference Review: ICLR (2022-2023), ICASSP (2022-2023), ICML (2020-2024), CHIL (2022), NeurIPS (2021-2022, 2024), ISIT (2022), ECML(2023), ICRA(2023).
- Workshops Review: IEEE Statistical Signal Processing Workshop (SSP 2021), Women in Machine Learning (2019), Women in Data Science and Mathematics (WiSDM 2023).

Presentations

Invited Talks

- Breaking the Curse of Multiagency in Robust Multi-Agent Reinforcement Learning.
2025 INFORMS Annual Meeting 2025.10
- Robustness: Requirement? An Opportunity — Robust Decision Making Without Compromising Data Efficiency.
Mitsubishi Electric Research Laboratories (MERL) Seminar Series 2026.1
UIUC, Department of Industrial and Enterprise Systems Engineering 2025.11
Tsinghua University, Institute for Interdisciplinary Data Science 2025.11
2025 SIAM New York-New Jersey-Pennsylvania Section Conference (SIAM NNP) 2025.10
John Hopkins Biostat Department Seminar Series. 2025.09
International Conference On Continuous Optimization (ICCOPT) 2025.07
Southern California Applied Mathematics Symposium 2025.04
Michigan RL Seminar series. 2025.04
The 44th Southern California Control Workshop at USC 2024.11
The Asilomar Conference on Signals, Systems, and Computers 2024.10
2024 INFORMS Annual Meeting 2024.10
- The Power of Distributional Robustness. *Georgia Tech ISyE Junior Researcher Workshop* 2024.4
- The Cost of Distributional Robustness in Reinforcement Learning: Minimax-Optimal Sample Efficiency.
Information Theory and Applications Workshop (ITA) 2024.2
Rising Stars In Machine Learning, University of Maryland 2023.11
WORDS 2023: Workshop in Operations Research and Data Science, Duke University 2023.11
Safe Reinforcement Learning Online Seminar 2023.10
2023 INFORMS Annual Meeting 2023.10
RSRG/FALCON Tea Talk at Caltech 2023.9
- Sample-Efficient Artificial Intelligence. *IST ADVISORY COUNCIL MEETING at Caltech* 2023.10
- Provable Algorithms for Reinforcement Learning: Efficiency and Robustness
UT Austin Rising Stars in Computational and Data Sciences 2023.4

- Information Theory and Applications Workshop (ITA)* 2023.2
- UChicago Rising Stars in Data Science (2022)* 2022.11
- Offline Reinforcement Learning: Towards Optimal Sample Complexity and Distributional Robustness.
Computer Science & Engineering at University of Connecticut 2023.2
Air Force Center of Excellence Meeting 2023.1
- Reinforcement learning based on game theory, stochastic composition optimization and regularization. *Intern presentation at Google Brain* 2022.9
- Manifold Gradient Descent Solves Multi-Channel Sparse Blind Deconvolution Provably and Efficiently.
SIAM Conference on Imaging Science (IS22) 2022.3
IEEE East Asian School of Information Theory (2021) 2021.8
Optimization interest seminar in Mitsubishi Electric Research Laboratories (MERL) 2020.7
- Fusion-Based Digital Image Correlation Framework for Strain Measurement. *Intern presentation at Mitsubishi Electric Research Laboratories (MERL)* 2020.8

Poster presentations at conferences/workshops

- Promoting Robustness Goes Beyond: Unlocking Data Efficiency, Computation Tractability, and Emergent Smarter solutions.
The 2024 Young Researchers Workshop at Cornell 2024.10
EECS Rising Stars Workshop at MIT 2024.10
- Provable Algorithms for Reinforcement Learning: Efficiency and Robustness.
2023 INFORMS Annual Meeting QSR Poster Competition Session 2023.10
ICASSP Rising Stars in Signal Processing 2023.6
- Curriculum Reinforcement Learning using Optimal Transport via Gradual Domain Adaptation. *Conference on Neural Information Processing Systems (NeurIPS),* 2022.12
- Pessimistic Q-Learning for Offline Reinforcement Learning: Towards Optimal Sample Complexity. *International Conference on Machine Learning (ICML),* 2022.7
- Breaking the Sample Complexity Barrier to Regret-Optimal Model-Free Reinforcement Learning. *Conference on Neural Information Processing Systems (NeurIPS),* 2021.12
- Robust camera pose estimation for image stitching. *IEEE International Conference on Image Processing (ICIP),* 2021.9
- Fusion-Based Digital Image Correlation Framework for Strain Measurement. *International Conference on Acoustics, Speech, and Signal Processing (ICASSP),* 2021.6
- Nonconvex Optimization for Multi-channel Sparse Blind Deconvolution. *Women in Machine Learning at NeurIPS,* 2019.12
- Device-free Multiple People Localization through Floor Vibration. *First ACM Workshop on Device-Free Human Sensing,* 2019.11

Full list of Honors & Awards

Honors/Fellowships:

- 2024 CMU ECE A.G. Milnes Award — the PhD thesis work judged to be of the highest quality and which has had or is likely to have significant impact in his or her field
- 2024 The Junior Researcher Workshop Invitation by Cornell

2024 Rising Stars In Electrical Engineering and Computer Science (EECS)
 2024 The ISyE Junior Researcher Workshop Invitation by Georgia Tech
 2023 Rising Stars In Machine Learning by University of Maryland
 2023 Computing, Data, and Society Postdoctoral Fellow by Caltech
 2023 ICASSP Rising Stars in Signal Processing
 2023 UT Austin Rising Stars in Computational and Data Sciences
 2022 UChicago Rising Stars in Data Science
 2022 NeurIPS 2022 Top Reviewer
 2022 Leo Finzi Memorial Fellowship
 2022 Wei Shen and Xuehong Zhang Presidential Fellowship
 2021 Liang Ji-Dian Graduate Fellowship
 2019 Women in Machine Learning Scholarship
 2018 Presidential Fellowship granted by Carnegie Mellon University
 2018 Carnegie Institute of Technology Dean's Fellowship
 2018 Excellent Honors Graduate granted by Tsinghua University
 2017 The First Prize in 35th Tsinghua University Academic Challenge Cup
 2015-2017 Technology Innovation Excellence Award granted by Tsinghua University
 2017 Enterprise Sponsored Scholarship granted by Tsinghua University
 2016 National Scholarship granted by the government of China
 2016 Qualcomm Scholarship granted by Tsinghua University
 2016 Outstanding Project of Undergraduate Research Competition of Tsinghua University
 2015 The First Prize in National Physics Contest for College Student
 2014 The Silver Medal of Chinese Physics Olympiad

Travel Awards:

2022 IAS WAM Mathematics in Machine Learning Travel Support
 2020 National Science Foundation (NSF) Student Travel Grants for ICASSP
 2020 Computing Research Association Grad Cohort for Women Travel Support
 2019 The ACM International Workshop on Device-Free Human Sensing Travel Grant
 2019 Women in Data Science and Mathematics Travel Support